

LONGUE-POINTE-DE-MINGAN, QC, Canada

WMO: 715120

Lat: 50.271N

Lon: 64.226W

Elev: 11

StdP: 101.19

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -27.2	(c) -24.4	(d) -31.1	(e) 0.2	(f) -26.0	(g) -28.5	(h) 0.3	(i) -23.3	(j) 8.4	(k) -5.1	(l) 7.4	(m) -5.4	(n) 1.9	(o) 340	(p) 0.671

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.6	(c) 21.6	(d) 16.5	(e) 20.2	(f) 16.0	(g) 18.9	(h) 15.4	(i) 17.9	(j) 20.0	(k) 17.0	(l) 18.8	(m) 16.2	(n) 18.0	(o) 2.8	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.1	12.2	18.8	16.2	11.5	18.0	15.4	10.9	17.1	50.8	19.9	48.0	18.8	45.6	18.0	22.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.8	(b) 5.9	(c) 5.2	(e) -32.2	(f) 25.7	(g) 2.6	(h) 2.5	(i) -34.1	(j) 27.5	(k) -35.7	(l) 28.9	(m) -37.1	(n) 30.3	(o) -39.1	(p) 32.1		
(a) 6.8	(b) 5.9	(c) 5.2	(e) -32.2	(f) 20.0	(g) 2.7	(h) 1.3	(i) -34.1	(j) 20.9	(k) -35.7	(l) 21.6	(m) -37.2	(n) 22.4	(o) -39.1	(p) 23.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.6	-12.5	-11.5	-6.4	0.4	5.5	10.6	14.0	13.7	9.4	4.6	-1.2	-8.3		
	DBStd	10.14	6.44	6.36	5.87	3.44	2.72	2.86	2.18	2.64	3.03	3.10	4.53	5.99		
	HDD10.0	3386	696	602	509	288	142	26	1	3	46	170	335	568		
	HDD18.3	6109	954	836	767	538	398	231	135	146	267	426	585	827		
	CDD10.0	321	0	0	0	0	3	45	125	117	29	2	0	0		
	CDD18.3	2	0	0	0	0	0	0	1	1	0	0	0	0		
	CDH23.3	11	0	0	0	0	0	1	6	4	1	0	0	0		
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0		
(14) Wind	WSAvg	2.4	2.5	2.6	2.7	2.7	2.3	2.3	2.1	2.0	2.2	2.4	2.5	2.6		
(15) Precipitation	PrecAvg	1118	84	68	89	78	92	93	96	86	98	117	121	100		
	PrecMax	1292	194	145	180	188	205	165	189	147	179	199	210	253		
	PrecMin	964	19	24	35	19	36	39	49	45	44	38	50	42		
	PrecStd	98	41	29	44	40	42	41	40	27	31	40	40	46		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.1	5.0	7.2	12.2	17.9	22.0	23.9	23.4	20.4	15.9	10.2	6.0		
		MCWB	2.6	3.6	2.9	5.9	11.5	15.0	16.9	17.7	15.6	12.7	9.5	5.3		
	2%	DB	1.2	2.5	4.7	9.0	14.6	19.6	21.8	21.4	17.8	12.9	7.9	3.8		
		MCWB	0.7	1.1	1.5	4.6	9.7	14.0	16.4	17.1	14.1	10.5	7.2	3.0		
	5%	DB	-0.2	0.4	2.9	7.0	12.8	17.8	20.3	20.0	16.2	11.1	6.3	2.2		
		MCWB	-0.5	-0.5	0.7	3.4	8.7	13.1	16.0	16.5	13.4	9.3	5.5	1.4		
	10%	DB	-2.2	-1.7	1.5	5.6	11.2	16.3	19.0	18.7	14.9	9.8	5.1	0.6		
		MCWB	-2.8	-2.6	-0.2	2.9	7.8	12.4	15.6	15.8	12.6	8.3	4.3	0.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.6	4.3	4.0	7.2	12.8	17.2	19.2	19.3	17.1	14.0	9.7	5.6		
		MCDB	3.4	4.8	6.0	10.8	15.0	19.9	21.8	21.8	18.8	14.9	9.9	5.8		
	2%	WB	0.7	1.1	2.4	5.6	10.7	15.4	17.9	18.1	15.3	11.2	7.4	3.2		
		MCDB	1.0	1.9	3.9	7.6	13.2	17.9	20.0	20.0	16.6	12.1	7.7	3.6		
	5%	WB	-0.8	-0.7	1.1	4.3	9.4	14.3	16.9	17.2	14.1	9.8	5.7	1.6		
		MCDB	-0.4	0.1	2.3	6.3	11.8	16.7	19.0	18.9	15.6	10.6	6.1	2.1		
	10%	WB	-3.0	-2.8	0.1	3.2	8.3	13.1	16.2	16.4	13.0	8.6	4.4	0.1		
		MCDB	-2.5	-2.1	1.4	5.1	10.7	15.3	18.1	18.1	14.3	9.5	5.0	0.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.7	11.2	10.9	8.7	9.7	10.0	9.6	9.8	9.9	8.2	7.5	9.1		
		MCDBR	9.4	8.8	9.3	11.1	13.1	12.8	12.3	11.2	11.5	9.9	7.2	6.9		
	5% WB	MCWBR	8.8	8.1	7.1	7.2	8.4	7.5	7.4	7.2	7.9	7.5	6.5	6.3		
		MCWBR	9.4	8.4	8.3	9.4	11.4	10.2	9.7	9.0	9.5	8.3	6.7	6.9		
(40) Clear-Sky Solar Irradiance	taub		0.266	0.270	0.293	0.338	0.362	0.384	0.417	0.381	0.355	0.330	0.291	0.273		
	taud		2.361	2.330	2.335	2.334	2.409	2.381	2.334	2.439	2.480	2.516	2.518	2.372		
	Ebn at Noon		786	878	917	906	895	874	836	853	841	801	760	717		
	Edh at Noon		63	84	101	114	111	115	119	102	88	72	55	55		
(44) All-Sky Solar Radiation	RadAvg		1.05	2.03	3.35	4.50	5.23	5.59	5.30	4.50	3.40	2.08	1.09	0.81		
	RadStd		0.12	0.17	0.29	0.38	0.32	0.51	0.49	0.23	0.22	0.15	0.10	0.07		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	+0.65	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	+1.24	N/A	+0.33	+0.36	N/A	N/A	N/A	N/A

Nomenclature: See separate page